7 KVADRATIK FORMALAR

1. $f(x_1;\ x_2;\ x_3) = 4x_1^2 - 12x_1x_2 - 10x_1x_3 + x_2^2 - 3x_3^2$ kvadratik formani matritsaviy ko‘rinishda yozing.

Yechish. Kvadratik formaning matritsasini topamiz:

$$f = (x_1\ \ x_2\ \ x_3)\begin{pmatrix} 4 & -6 & -5 \\ -6 & 1 & 0 \\ -5 & 0 & -3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad A = \begin{pmatrix} 4 & -6 & -5 \\ -6 & 1 & 0 \\ -5 & 0 & -3 \end{pmatrix}$$ ▶

? Kvadratik formalarni matritsaviy ko‘rinishda yozing.

2. $f = -3x_1^2 - 4x_1x_2 - 6x_1x_3 + 8x_2^2 - x_3^2$

3. $f = x_1^2 - 6x_1x_2 - 8x_2x_3 + 14x_2^2 - 5x_3^2$

4. $f(x_1;\ x_2) = 2x_1^2 + 4x_1x_2 - 3x_2^2$ kvadratik forma berilgan. $\begin{cases} x_1 = 2y_1 - 3y_2 \\ x_2 = y_1 + y_2 \end{cases}$ chiziqli almashtirish orqali hosil bo‘lgan $f(y_1;\ y_2)$ kvadratik formani toping.

Yechish. Berilgan kvadratik formaning matritsasi $A = \begin{pmatrix} 2 & 2 \\ 2 & -3 \end{pmatrix}$ chiziqli almashtirish matritsasi $C = \begin{pmatrix} 2 & -3 \\ 1 & 1 \end{pmatrix}$ bo‘ladi. Qidirilayotgan kvadratik formaning matritsasini quyidagicha topamiz:

$$A' = C^t \cdot A \cdot C = \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 2 & -3 \end{pmatrix}\begin{pmatrix} 2 & -3 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 13 & -17 \\ -17 & 3 \end{pmatrix}$$

kvadratik formaning ko‘rinishi: $f(y_1;\ y_2) = 13y_1^2 - 34y_1y_2 + 3y_2^2$ ▶

? Mashqlarni bajaring.

5. $f = 3x_1^2 - 6x_1x_2 - x_2^2$

kvadratik forma berilgan

$$\begin{cases} x_1 = -y_1 - y_2, \\ x_2 = -3y_1 + 6y_2, \end{cases}$$

almashtirish yordamida hosil bo‘lgan kvadratik formani toping.

6. Kvadratik forma $f(x_1; x_2) = 3x_1^2 - 6x_2^2 + 14x_1x_2$, berilgan. $\begin{cases} x_1 = 2y_1 - 3y_2, \\ x_2 = 5y_1 + 3y_2, \end{cases}$

chiziqli almashtirish orqali hosil bo‘lgan kvadratik formani toping.

7. Kvadratik formani kanonik ko‘rinishga keltiring.

$$f(x_1; x_2; x_3) = x_1^2 - 3x_1x_2 + 4x_1x_3 + 2x_2x_3 + x_3^2$$

Yechish. $f(x_1; x_2; x_3) = x_1^2 - x_1(3x_2 - 4x_3) + 2x_2x_3 + x_3^2$

x_1 o‘zgaruvchining kvadrati o‘rnida turgan koeffitsiyenti no‘ldan farqli bo‘lgani uchun, x_1 o‘zgaruvchining to‘liq kvadratini topamiz:

$$f = \left[x_1^2 - 2x_1\left(\frac{1}{2}(3x_2 - 4x_3)\right) + \left(\frac{1}{2}(3x_2 - 4x_3)\right)^2\right] - \left(\frac{1}{2}(3x_2 - 4x_3)\right)^2 + 2x_2x_3 + x_3^2 =$$

$$\left(x_1 - \frac{3}{2}x_2 + 2x_3\right)^2 - \frac{9}{4}x_2^2 + 8x_2x_3 - 3x_3^2$$

endi o‘zgaruvchi x_2 uchun kvadratinit opamiz:

$$f = \left(x_1 - \frac{3}{2}x_2 + 2x_3\right)^2 - \frac{9}{4}\left(x_2 - \frac{16}{9}x_3\right)^2 + \frac{37}{9}x_3^2,$$

Demak, no‘ldan farqli chiziqli almashtirish

$$y_1 = x_1 - \frac{3}{2}x_2 - 2x_3 \; y_2 = x_2 - \frac{16}{9}x_3 \; y_3 = x_3$$

berilgan kvadratik formani kanonik ko‘rinishga keltiradi:

$$f(y_1, y_2, y_3) = y_1^2 - \frac{9}{4}y_2^2 + \frac{37}{9}y_3^2$$

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Kvadratik formani kanonik ko‘rinishga keltiring.

8. $f(x_1,x_2,x_3)=x_1^2+12x_1x_2-15x_1x_3+5x_2^2-3x_3^2$

9. $f(x_1,x_2,x_3)=3x_1^2+13x_1x_2-5x_1x_3+12x_2^2-13x_3^2$

10. $f(x_1,x_2,x_3)=3x_1^2+2x_1x_2-5x_1x_3+3x_2^2-13x_3^2$

11. $f(x_1,x_2,x_3)=15x_1^2+9x_1x_2-15x_1x_3+12x_2^2-6x_3^2$

12. $f(x_1,x_2,x_3)=4x_1^2+12x_1x_2-5x_1x_3+13x_2^2-3x_3^2$

13. $f(x_1,x_2,x_3)=3x_1^2+3x_1x_2-5x_1x_3+12x_2^2-3x_3^2$

14. $f(x_1,x_2,x_3)=3x_1^2+12x_1x_2-15x_1x_3+3x_2^2-13x_3^2$

15. $f(x_1,x_2,x_3)=3x_1^2+12x_1x_2-15x_1x_3+3x_2^2-13x_3^2$

16. $f(x_1;x_2)=13x_1^2-6x_1x_2+5x_2^2$ kvadratik formaning ishorasini aniqlang.

Yechish. Kvadratik formaning matritsasi $A=\begin{pmatrix}13 & -3\\ -3 & 5\end{pmatrix}$ bo‘ladi.

Xarakteristik tenglama tuzamiz: $|A-\lambda E|=\begin{vmatrix}13-\lambda & -3\\ -3 & 5-\lambda\end{vmatrix}=0$

yoki $\lambda^2-18\lambda+56=0$ ya’ni $\lambda_1=14,\quad \lambda_2=4$ xarakteristik tenglamaning yechimlari musbat bo‘lgani uchun, f - musbat aniqlangan kvadratik forma bo‘ladi. ▶

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Quyidagi kvadratik formalarning ishoradasi aniqlang:

17. $f(x_1;x_2;x_3)=2x_1x_2+x_2^2-2x_3^2+4x_1x_3-2x_2x_3$

18. $f(x_1;x_2;x_3)=7x_1^2-4x_1x_2+3x_2^2-x_3^2+8x_2x_3$

19. $f(x_1;x_2;x_3)=-3x_1^2-6x_1x_2+x_3^2+10x_1x_3-8x_2x_3$

20. $f(x_1;x_2;x_3)=2x_1^2-14x_1x_2+x_2^2-x_3^2$

21. $f(x_1;x_2;x_3)=x_1^2-x_2^2-6x_3^2+12x_1x_2$

22. $f(x_1;x_2;x_3)=-x_2^2-x_1^2-8x_1x_3+2x_2x_3-2x_3^2$

23. $f(x_1;x_2;x_3)=5x_1^2+36x_2^2+12x_1x_2$

24. $f(x_1;x_2;x_3)=2x_1^2-4x_1x_2-x_2^2+15x_3^2$

25. $f(x_1;x_2;x_3)=x_1x_2+4x_2^2+3x_3^2+4x_1x_3-6x_2x_3$

26. $f(x_1;x_2;x_3)=x_1^2-14x_1x_2+3x_2^2-5x_3^2+4x_2x_3$

27. $f(x_1;x_2;x_3)=x_1^2-4x_1x_2+12x_3^2+8x_1x_3-8x_2x_3$

28. $f(x_1;x_2;x_3)=2x_1^2-22x_1x_2+x_2^2+15x_3^2$

29. $f(x_1;x_2;x_3)=-6x_1^2+4x_2^2+3x_3^2+24x_1x_2$

30. $f(x_1;x_2;x_3)=12x_2^2-x_1^2+3x_1x_3+2x_2x_3-2x_3^2$

31. $f(x_1;x_2)=9x_1^2+6x_2^2+10x_1x_2$

32. $f(x_1;x_2;x_3)=2x_1^2-4x_1x_2+5x_2^2+5x_3^2$

Foydalanishga tavsiya etiladigan adabiyotlar ro‘yxati

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